

Frame transport for framed formal monodromy: the lemma, its hypotheses, and the receiver problem

Second version, correcting the first

Memo for a reader of *Irrationality of Cubic Threefolds after One Stabilization*

15 August 2026

Abstract

Proposition 4.7 of the draft transports framed formal monodromy across the comparison isomorphisms of Iritani and Iritani–Koto, and the transport step itself is asserted in one sentence. This memo states and proves the missing lemma over a differential field, with its hypotheses made explicit; shows that those hypotheses are jointly unsatisfiable over the coefficient receiver the draft builds, so that the lemma as proved has no verified instance in the application; corrects the receiver; and reduces the remaining question to one inequality that depends on a normalization the draft is free to choose. The inequality fails under the draft’s current instruction to take the separating weight parameter large, and holds at the cubic endpoint under the smallest admissible choice. Section 4 examines an iterated receiver that would satisfy the inequality trivially, checks it against the sources, and finds that it needs a semipositivity hypothesis on the ambient variety which weak factorization does not supply. Section 5 then identifies what is probably the actual repair, in the framing theory of Hinault, Yu, Zhang and Zhang, whose decomposition gauge is regular in the loop coordinate and so satisfies the transport lemma’s hypothesis outright. Section 8 lists what the first version of this memo got wrong; readers of that version should start there.

1 The assertion at issue

References are to Section 4 of the draft, *Framed formal monodromy*. I name statements by their semantic labels as well as by number, because the edits proposed in Section 7 renumber the section: Definition 4.1 (`def:framed-sixth-multiplicity`), Definition 4.2 (`def:pro-laurent-gauge-group`), Lemma 4.3 (`lem:pv-base-change`), Lemma 4.5 (`lem:formal-base-shift`), Proposition 4.7 (`prop:framed-operations`), Lemma 4.9 (`lem:divisor-tagging`).

Lemma 4.3 ends with the sentence

If the gauge uses only integral powers of z at every finite level, this conjugacy preserves the original- z -disc frame,

whose proof is the single sentence “the deck action fixes every integral power of z at each finite level, so the compatible gauge is single-valued for the framed turn.” Lemma 4.5 then *defines*

$$M^{\text{bulk}} = G M_{\mathfrak{f},V}^{\text{small,shifted}} G^{-1} \tag{4.1c}$$

and Proposition 4.7 concludes that “each center target summand has the framed monodromy of its specialized small connection.” The conclusion does not follow from the display: conjugation

preserves a characteristic polynomial for trivial reasons, but says nothing about whether the conjugated matrix is the intrinsic turn operator on horizontal solutions of the gauged module. What is missing is a transport lemma together with an algebra in which the turn acts on solutions.

2 The lemma over a differential field

Notation for Exp_V , $K = \mathbf{C} \oplus V$ and Ω_V is the draft's, from (4.0) and (4.0a). Recall that Exp_V is defined on $(K, +)$, not on Ω_V .

The ground field

Let Γ be a totally ordered abelian group and let

$$\mathcal{H} = \Omega_V((\Gamma \oplus \mathbf{Z}e_z))$$

be the Hahn field of series with well-ordered support and coefficients in Ω_V , the value group ordered lexicographically with the e_z -coordinate *last*, as in the draft. Write $z = x^{e_z}$ and let θ multiply the coefficient at (γ, k) by k ; this is a derivation because k is additive on the value group, and $\theta z = z$. Its constants are

$$\mathcal{C} := \mathcal{H}^{\theta=0} = \Omega_V((\Gamma)).$$

Taking $\Gamma = 0$ gives $\mathcal{H} = \Omega_V((z))$. For $e \geq 1$ put $\mathcal{H}_e = \Omega_V((\Gamma \oplus \frac{1}{e}\mathbf{Z}e_z))$, a degree- e extension containing $w = x^{e_z/e}$, and $\theta_w = e\theta$, so $\theta_w w = w$. All residues below are taken with respect to θ_w on the ramified side; the draft's $M_{\text{RS},V}$ uses the same convention, and the two differ by the factor e if one is careless.

The solution algebra and the turn

Definition 1. Assume the exponential factors under consideration have *constant* coefficients, that is $\varphi \in w^{-1}\Omega_V[w^{-1}]$. The *universal formal solution algebra* $\mathcal{U}_e$ is the $\mathcal{H}_e$ -algebra generated by symbols

$$w^\rho \ (\rho \in K), \quad \ell, \quad E(\varphi),$$

subject to $w^\rho w^{\rho'} = w^{\rho+\rho'}$, w^n = the element w^n of $\mathcal{H}_e$ for $n \in \mathbf{Z}$, $E(\varphi)E(\varphi') = E(\varphi + \varphi')$, $E(0) = 1$, with

$$\theta_w(w^\rho) = \rho w^\rho, \quad \theta_w(\ell) = 1, \quad \theta_w(E(\varphi)) = \theta_w(\varphi)E(\varphi).$$

The two hypotheses buried in that definition — constant coefficients in φ , and exponents ρ in K rather than in Ω_V — are not bookkeeping. The first is false for the modules the draft transports and is discussed at Remark 5; the second is forced because Exp_V is defined only on K , so a module whose residues leave K needs Exp rebuilt over a larger field.

Lemma 2 (Structure). *Fix representatives $\Lambda \subset K$ for $K/\mathbf{Z}$ with $0 \in \Lambda$. Then $\mathcal{U}_e$ is free as an $\mathcal{H}_e$ -module with basis $\{w^\rho \ell^m E(\varphi)\}$, $\rho \in \Lambda$, $m \geq 0$. In particular $\mathcal{U}_e \neq 0$ and $\mathcal{H} \subset \mathcal{U}_e$.*

Proof. One cannot argue that $K/\mathbf{Z}$ is torsion-free; it is not, since it contains $\mathbf{Q}/\mathbf{Z}$, and the rational classes are exactly the ones ν_6 counts. Instead use base change. The group algebra $\mathcal{H}_e[K]$ is free over $\mathcal{H}_e[\mathbf{Z}] = \mathcal{H}_e[t, t^{-1}]$ on any set Λ of coset representatives, and $\mathcal{U}_e$ is obtained from it by the base change $\mathcal{H}_e[t, t^{-1}] \rightarrow \mathcal{H}_e$, $t \mapsto w$, a surjection with kernel $(t - w)$ generated by a nonzerodivisor. Base change of a free module is free on the image of the same basis. The crossed multiplication $w^\rho w^{\rho'} = w^n w^{\rho''}$ is well defined, and the derivation is consistent with the integer identification because w^n sits at e_z -exponent n/e and θ_w multiplies it by n . The factors $E(\varphi)$ form the group algebra of a torsion-free group and ℓ is a polynomial variable. $\square$

Lemma 3 (The turn). *There is a unique automorphism σ of $\mathcal{U}_e$ with*

$$\sigma(w) = e^{2\pi i/e} w, \quad \sigma(w^\rho) = \text{Exp}_V(\rho/e) w^\rho, \quad \sigma(\ell) = \ell + \frac{2\pi i}{e}, \quad \sigma(E(\varphi)) = E(\sigma\varphi),$$

acting on $\mathcal{H}_e$ by multiplying the coefficient at $(\gamma, j/e)$ by $e^{2\pi i j/e}$. It fixes $\mathcal{H}$, hence $\mathcal{C}$, pointwise, and commutes with θ_w .

Proof. The action on $\mathcal{H}_e$ multiplies coefficients by a character of the value group, hence is a support-preserving ring automorphism, and is the identity on $\mathcal{H}$ because there $j \in e\mathbf{Z}$. Multiplicativity in ρ is the homomorphism property of Exp_V on $(K, +)$, and the two prescriptions agree on integral exponents because $\text{Exp}_V(n/e) = e^{2\pi i n/e}$ by (4.0a). Commutation with θ_w is not a diagonality argument, since neither $\sigma(\ell) = \ell + 2\pi i/e$ nor $\theta_w(\ell^m) = m\ell^{m-1}$ is diagonal; it is the direct check $\sigma\theta_w(\ell^m) = m(\ell + \frac{2\pi i}{e})^{m-1} = \theta_w\sigma(\ell^m)$, together with the diagonal cases. $\square$

Lemma 4 (Constants). *Under the constant-coefficient hypothesis of Definition 1, $\mathcal{U}_e^{\theta_w=0} = \mathcal{C}$.*

Proof. Let v be the Hahn valuation of $\mathcal{H}_e$, additive because the coefficient field is a field; θ_w multiplies coefficients by integers and never enlarges a support, so $v(\theta_w c) \geq v(c)$.

Write $x = \sum c_{\rho, m, \varphi} w^\rho \ell^m E(\varphi)$ in the basis of Lemma 2 with $\theta_w x = 0$; the equation splits over φ . For $\varphi \neq 0$, let M be the top ℓ -degree in that block; for each ρ , $\theta_w(c_{\rho, M}) + \rho c_{\rho, M} + \theta_w(\varphi)c_{\rho, M} = 0$. Every monomial of $\theta_w(\varphi)$ has Γ -part 0 and strictly negative e_z -part, so $v(\theta_w(\varphi)c) < v(c) \leq v(\theta_w c + \rho c)$ for $c \neq 0$ and the lowest term cannot cancel; hence $c_{\rho, M} = 0$, and descending in M kills the block.

For $\varphi = 0$ the coefficient equations are $\theta_w(c_{\rho, m}) + \rho c_{\rho, m} + (m+1)c_{\rho, m+1} = 0$. Descending in m : the top-degree coefficient satisfies the homogeneous equation $\theta_w(c) + \rho c = 0$, and on each monomial this reads $(j + \rho)c = 0$ with $j \in \mathbf{Z}$, so $c = 0$ unless $\rho \in \mathbf{Z}$, that is unless $\rho = 0$. For $\rho = 0$ the top coefficient is θ_w -constant and the next equation gives $M c_{0, M} = -\theta_w(c_{0, M-1})$, whose right side has zero component in e_z -degree 0; comparing that component gives $c_{0, M} = 0$ for $M \geq 1$. What survives is $\rho = 0$, $m = 0$, $\varphi = 0$ with θ_w -constant coefficient, and the θ_w -constants of $\mathcal{H}_e$ are those supported in e_z -degree zero, namely $\mathcal{C}$. $\square$

Remark 5 (The constant-coefficient hypothesis is necessary, and fails in the application). Allow φ a Novikov coefficient of positive weight, $\varphi = x^\gamma w^{-1}$ with $\gamma \in \Gamma$, $\gamma > 0$ — the shape of every exponential factor the draft actually meets, since the eigenvalues λ of $c_1 \star$ have positive Novikov weight. Then $v(\theta_w \varphi) > 0$ in the coefficient-dominant order and the domination argument reverses. Worse, the equation $\theta_w(c) = \theta_w(\varphi)c$ is solved by

$$c = \sum_{n \geq 0} \frac{(-1)^n}{n!} x^{n\gamma} w^{-n},$$

whose support $\{(n\gamma, -n/e)\}_{n \geq 0}$ is increasing, hence well ordered, so $c \in \mathcal{H}_e$ and $cE(\varphi)$ is a θ_w -constant outside $\mathcal{C}$. Lemma 4 is therefore false in that regime, and with it Proposition 6. The structural reason is worth stating on its own: *in the draft's coefficient-dominant order, $\exp(-\lambda/z)$ is already a unit of the coefficient field*, so adjoining $E(\varphi)$ as a free symbol duplicates an existing element and inflates the constants. In that order the exponential factors are not symbols at all, and the entire irregularity of the formal solution sits in its divergent unit part, which is exactly where the rate comparison of Section 3 lives.

Framed operators and transport

Proposition 6 (The framed operator is the matrix of the turn). *Let D be a differential module over $\mathcal{H}$, free of rank n , with connection matrix Ω , and let $Y \in \mathrm{GL}_n(\mathcal{U}_e)$ satisfy $\theta Y = -\Omega Y$. Then $\mathrm{Sol}(D) = \{v : \theta v + \Omega v = 0\} = Y\mathcal{C}^n$, free of rank n and σ -stable; the matrix M with $\sigma(Y) = YM$ lies in $\mathrm{GL}_n(\mathcal{C})$ and is well defined up to $\mathrm{GL}_n(\mathcal{C})$ -conjugacy. For D over $\Omega_V((z))$ in Levelt–Turrittin form, M is the draft's framed operator: in a solution basis adapted to one deck orbit of length d the matrix of σ is block-cyclic with the draft's intertwiners A_i as blocks, and the cyclic-block determinant gives $\chi(X) = \det(X^d I - U)$ with $U = A_{d-1} \cdots A_0$, which is (4.0b).*

Proof. From $\theta(Y^{-1}) = Y^{-1}\Omega$ one gets $\theta(Y^{-1}v) = Y^{-1}(\theta v + \Omega v)$, so horizontality of v means $Y^{-1}v$ has θ_w -constant entries, and Lemma 4 identifies those with $\mathcal{C}$. Since σ commutes with θ and fixes Ω , it preserves $\mathrm{Sol}(D)$; so $M = Y^{-1}\sigma(Y)$ has entries in $\mathcal{C}$ and is invertible, being a product of invertibles, with inverse $\sigma(Y^{-1})Y$. Two solution matrices differ on the right by an element of $\mathrm{GL}_n(\mathcal{C})$, which conjugates M . $\square$

Theorem 7 (Frame transport). *Let D, D' be as in Proposition 6, with solution matrices Y, Y' , and let $G \in \mathrm{GL}_n(\mathcal{H})$ satisfy $\Omega' = G\Omega G^{-1} - \theta(G)G^{-1}$. Then GY is a solution matrix for D' and $\sigma(GY) = (GY)M$. Consequently*

$$M_{f,V}(D') = C_0^{-1} M_{f,V}(D) C_0 \quad \text{for some } C_0 \in \mathrm{GL}_n(\mathcal{C}),$$

so the two framed operators have the same characteristic polynomial in $\mathcal{C}[X]$ and the same multiplicity at every root of unity.

Proof. $\theta(GY) = \theta(G)Y - G\Omega Y = -\Omega'(GY)$. Since σ fixes $\mathcal{H}$ pointwise, $\sigma(G) = G$, so $\sigma(GY) = G\sigma(Y) = GYM$. Writing $Y' = (GY)C_0$ with $C_0 \in \mathrm{GL}_n(\mathcal{C})$ gives $M' = C_0^{-1}MC_0$. $\square$

Remark 8 (Which object is $M_{f,V}$). The framed operator is an automorphism of the horizontal module; its matrix depends on a choice of *solution* frame and is well defined up to conjugacy over the constants. The conjugate GMG^{-1} appearing in the draft's (4.1c) is not the matrix of the turn in any solution frame — Y' differs from GY by a constant matrix, not by G — but the matrix of the transported automorphism read in the *module* frame of D' . Both readings are legitimate, and they give the same characteristic polynomial, but the draft should say which it means, since conflating them is exactly the step at issue.

Remark 9 (The hypothesis on G is the whole content). Since σ fixes $\mathcal{H}$ pointwise, $\sigma(G) = G$ is automatic once G has entries in $\mathcal{H}$; what matters is that G lies in $\mathcal{H}$ and not in $\mathcal{U}_e$. No hypothesis-free statement exists: Y itself is a gauge over $\mathcal{U}_e$ carrying D to the trivial module, whose framed operator is the identity. The rank-one case shows what survives: if $g \in \mathcal{H}^\times$ and $\theta(g)/g = \delta$, then each monomial gives $(k - \delta)g_{(\gamma,k)} = 0$, so $\delta \in \mathbf{Z}$; a gauge over $\mathcal{H}$ shifts residues only by integers, which Exp_V kills.

The scissors: these hypotheses have no common instance in the application

Theorem 7 is true, and in the application it is empty. Its two hypotheses pull in opposite directions.

- If $\Gamma = 0$, so $\mathcal{H} = \Omega_V((z))$, then Levelt–Turrittin supplies solution matrices, but the pro-Laurent comparison gauge has unbounded negative z -order and is not in $\mathrm{GL}_n(\mathcal{H})$.
- If $\Gamma \neq 0$ with the draft’s coefficient-dominant order, the gauge lies in $\mathcal{H}$, but $\mathcal{H}$ is not a formal Laurent series field over its constants and Levelt–Turrittin is unavailable. Concretely, the e_z -component of the Hahn valuation is not a valuation on $\mathcal{H}$: for $f = x^{(0,0)} + x^{(1,-100)}$ and $g = -x^{(0,0)}$ its values on f and g are 0 while its value on $f + g$ is -100 . So there is no Newton polygon, no slope filtration and no Turrittin algorithm. Nor can one descend to the bounded- z -order elements: $f = 1 - zx^{-\gamma}$ has inverse $-\sum_{k \geq 1} z^{-k} x^{k\gamma}$, so they do not form a field.

An abstract Picard–Vessiot ring for a module over $\mathcal{H}$ does exist, with constants $\mathcal{C}$ after passing to the divisible hull. So the missing ingredient is *not* a solution algebra. It is a canonical Levelt–Turrittin normal form over $\mathcal{H}$, singling out which element of the differential Galois group deserves to be called the turn. That is the precise shape of the remaining problem, and it also shows why a purely valuation-theoretic argument cannot supply it.

3 The receiver

The draft’s receiver, in the proof of Proposition 4.7, is

$$\mathcal{R}_j = \Omega_{V,j} \otimes_{H_{0,j}} H_{\mathrm{tot},j}, \quad H_{\mathrm{tot},j} = \mathbf{C}((\Gamma_j^{\mathrm{coeff}} \oplus \mathbf{Z}e_z)). \quad (4.4c)$$

Why the tensor product cannot simply be dissolved

The map $a \otimes h \mapsto a \cdot h$ into the Hahn field with $\Omega_{V,j}$ coefficients is not well defined: the copy of $H_{0,j}$ inside $\Omega_{V,j}$ consists of degree-zero constants, the copy inside $H_{\mathrm{tot},j}$ of monomial series, and the tensor product exists precisely to identify them. Nor can the two copies be assigned different jobs, with module coefficients read as constants and gauge bookkeeping as monomials: the gauge relation $\Omega' = G\Omega G^{-1} - \theta(G)G^{-1}$ ties G and Ω through the same Novikov variables, so one copy is forced. With the constants reading the gauge leaves the field; with the monomial reading the exponential factors become Γ -supported and Remark 5 applies.

Two things should also be said about $\mathcal{R}_j$ itself. It is a tensor product of two field extensions and need not be a domain, and the draft reads characteristic polynomials and algebraic multiplicities in $\mathcal{R}_j[X]$; that deserves an explicit remark, which the draft partly gives by proving each factor injects.

A better receiver

Extend the Γ -valuation of $H_{0,j}$ to its algebraic closure and then to $\Omega_{V,j}$, and take the field of series $\sum_k c_k z^k$ with $c_k \in \Omega_{V,j}$ whose support $\{(v_\Omega(c_k), k)\}$ is well ordered in the chosen order. This has a single copy of the coefficients, is a field rather than a tensor product of two fields, hosts the pro-Laurent gauge, and hosts the exponential factors as honest units. It is strictly better than (4.4c). It does not by itself supply Levelt–Turrittin, and it does not remove the rate criterion below.

The two rates

Weight a monomial of coefficient weight d and z -power k by $a d + b k$, $a, b > 0$, and let ε be the minimal weight of a generator of J_j .

- *Pro-Laurent gauges gain one filtration unit per unit of z -pole.* By the recursion (4.1a) and the draft's own conclusion, G_α has no power of z below $-|\alpha|$ and a term of bulk degree m lies in $F^m B$; every monomial of J_j^N has weight at least N . Their weights grow like $m(a-b)$, which together with the draft's finite-below argument gives well ordered support when $a > b$.
- *Splitting gauges lose $w(\Delta\lambda)$ per unit of z -power.* This is the draft's own construction (4.9d): at order n the block off-diagonal coefficient is removed by a Sylvester equation $[K_0, A_n] = B_n$ with K_0 the leading matrix, so one division by an eigenvalue difference per order, and $w(A_n) \geq -n w(\Delta\lambda)$. The draft's displayed cubic entry $-14/(81r^2)$ at z -order two exhibits the rate: one factor of r^{-1} per unit of z -power. Three caveats. After a ramification $z = w^e$ the divisions occur once per w -order, so the loss is $e w(\Delta\lambda)$ per unit of z -power; at the cubic the draft proves the exponential blocks are unramified in z , so $e = 1$ there. The claim that the regular-singular normalization costs nothing needs the homogeneity argument that residue eigenvalues are rational, verified in the cubic block where $\det L_s = (s + 1/6)(s + 5/6)$. And $w(\Delta\lambda)$ is not controlled by $w(\lambda)$ when the Picard rank exceeds one, since w and the quantum grading are different functionals and leading terms can cancel.

Both conditions hold together iff

$$e w(\Delta\lambda) < \varepsilon$$

The criterion is not an artifact of choosing an order

For $C := e w(\Delta\lambda)/\varepsilon \geq 1$ the obstruction is not about well ordering at all. The z^n -coefficient of the single product GP of a pro-Laurent gauge with a splitting gauge is $\sum_m g_m p_{n+m}$, whose terms have weight at least $m(1-C) - nC$; for $C \geq 1$ infinitely many terms have bounded weight, so that coefficient is an infinite sum in no completion, ordered or not. Any repair that forms this product must therefore establish $C < 1$. A repair that never forms it — see route 2 below — is not bound by this.

The criterion depends on a normalization the draft may choose

This is the practical point, and it reverses the recommendation of the first version of this memo. The draft fixes $w(u) = 1$ and $w(s_{j,\ell}) = 1$ independently of L , and sets $w(Q^{i_* d} u^{\rho_C \cdot d}) = L(H \cdot i_* d) + \rho_C \cdot d$. Hence $\varepsilon = 1$ always, while the weights of the Novikov generators grow linearly in L . The criterion is therefore

$$e(L(H \cdot i_* d_0) + \rho_C \cdot d_0) < c_1 \cdot d_0,$$

using that the eigenvalues of $c_1 \star$ are homogeneous of quantum degree one when Q^d has degree $c_1 \cdot d$ — a convention I have checked against the draft's own low-dimensional vanishing proof, where the Q^β -component of $c_1 \star$ maps H^i to $H^{i+2-2c_1 \cdot \beta}$, and against its explicit cubic block, where K_0 has eigenvalues $\pm 6r, 0, 0$ with $r = (3q)^{1/2}$ and $c_1 \cdot \ell = 2$. Consequences:

- Under the draft’s instruction to choose L “so large that” separation holds, the criterion fails at essentially every comparison with two distinct exponential factors — including the cubic endpoint.
- Under the smallest admissible L it can hold. For a projective bundle with $c_1(V) \cdot \ell = 0$ one may take $L = 1$, giving $w(Q^\ell) = 1 = \varepsilon$ and $w(\Delta\lambda) = w(q^{1/2}) = 1/2$.
- Independently of L : separation forces $w \geq \varepsilon$ on every generator of J_j , and the image of Q^{d_0} is a generator, so $c_1 \cdot d_0 \geq 2$ is necessary under any admissible weight. Index one fails outright.

So the cheapest available repair is to choose the separating weight minimally rather than large, and to check the resulting inequality case by case. The first version of this memo asserted the opposite, on the false ground that the ratio was scale-invariant; rescaling w does move both rates equally, but L is not a rescaling, because the bulk generators’ weights are pinned at one.

4 The iterated receiver

The criterion above is a consequence of one architectural choice, and a reader of the second version of this memo proposed the alternative: the draft puts the intrinsic Novikov variables of the endpoint and the new completion variables into a single ordered group, which is what gives an eigenvalue difference positive weight. Nothing forces that. Treat the intrinsic Novikov fraction field as *coefficients*, and complete only in the new exceptional and bulk directions afterwards.

Two facts make the proposal concrete, and both are visible in the draft.

- *The unbounded negative z -order is bulk-driven, not Novikov-driven.* In the recursion (4.1a) the coefficient G_α has no power of z below $-|\alpha|$, where α indexes the *bulk* parameter η , and the filtration depth is acquired by substituting $\eta \in F^1 B$. A completion in the bulk directions alone therefore already hosts the pro-Laurent gauge; the Novikov directions are not needed for it.
- *With the Novikov directions as coefficients, the pathology disappears and the criterion is satisfied rather than evaded.* Eigenvalues of $c_1 \star$ become constants of weight zero, so $\exp(-\lambda/z)$ is not a unit of the receiver but a genuine exponential symbol, the constant-coefficient hypothesis of Definition 1 holds, and Remark 5 evaporates. Since $w(\Delta\lambda) = 0$, the inequality $e w(\Delta\lambda) < \varepsilon$ holds trivially, and the product GP of Section 3 converges coefficientwise.

The shifts that do involve the exceptional Novikov root do not force those directions back into the filtration. The rank-one unit twist attached to $-(c-1)\lambda_j$, the fixed divisor shift, and the ambient target’s $\tau^\circ = O(q^{-1})$ all enter Lemma 4.5 as its a_0 and a_2° terms, which are required only to be *constant in* the positive filtration, not to lie in it. With the Novikov directions as coefficients, $\exp(-a_0/z)$ is an honest exponential symbol rather than the object the draft must argue about separately.

What the sources actually permit

I have checked this against the pinned sources rather than left it as a question, and the answer is positive for the comparison and negative for the gauge.

Iritani's Theorem 5.18 states Ψ as an isomorphism of $\mathbf{C}[z]((q^{-1/s}))[[Q, \tilde{\tau}]]$ -modules, with the change of variables defined over $\mathbf{C}((q^{-1/(r-1)}))[[Q]]$. So the ring is *Laurent* in the exceptional root and *formal* in the ambient Novikov variables and the bulk. Two consequences.

The comparison survives. Inverting the Novikov monomials and completing in the exceptional and bulk directions is a ring homomorphism out of that ring. Ψ , its inverse, and the two inverse identities are module maps over it, so they base-change along it unchanged, and the framed operators' entries lie in the coefficient field, which injects. Nothing in the transport step needs Iritani's proof, only his statement, so the formality of his ring in Q is not by itself an obstacle.

The gauge does not. The source's own reconstruction writes $\tau(\tilde{\tau}) = \tau^\circ + t(\tilde{\tau})$ and $\varsigma_j(\tilde{\tau}) = \varsigma_j^\circ + s_j(\tilde{\tau})$, with $t(\tilde{\tau}) \in H^*(X)((q^{-1}))[[Q, \tilde{\tau}]]$ and s_j of the same shape over $H^*(Z)$. The initial terms are harmless: τ° and ς_j° are by definition the values at $Q = \tilde{\tau} = 0$, hence carry no Novikov dependence at all, and the exceptional direction remains a completion direction as the draft already has it. But t and s_j vanish only at $Q = \tilde{\tau} = 0$, so at $\tilde{\tau} = 0$ they retain pure-Novikov terms. Those have to be removed by the normalized gauge of Lemma 4.5, and that construction converges by topological nilpotence of the bulk shift. Inverting the Novikov directions destroys exactly that.

So the iterated receiver is not free, and the decisive question is now small and graded rather than general:

Is the pure-Novikov part of the coordinate shift — t and s_j restricted to $\tilde{\tau} = 0$ — confined to $H^0 \oplus H^2$?

If it is, the string and divisor equations absorb it into the a_0 and a_2° terms of Lemma 4.5, which are required only to be constant in the positive filtration, and the iterated receiver goes through for the ambient summand with $w(\Delta\lambda) = 0$. If it is not, the Novikov directions must stay inside the filtration, eigenvalue differences reacquire positive weight, and the criterion of Section 3 returns.

The grading answers this, and the answer is negative in general. Iritani's Theorem 5.18(5) states that $\tau(\tilde{\tau})$ and $\varsigma_j(\tilde{\tau})$ are homogeneous of degree 2, and his conventions are $\deg Q^d = 2c_1(X) \cdot d$ and $\deg \tau_i = 2 - \deg \phi_i$. A pure-Novikov term $Q^d \phi$ therefore satisfies

$$\deg \phi = 2 - 2c_1(X) \cdot d,$$

so $c_1 \cdot d = 1$ contributes in H^0 , $c_1 \cdot d = 0$ in H^2 , and $c_1 \cdot d = -1, -2, \dots$ in $H^4, H^6, \dots$; classes with $c_1 \cdot d \geq 2$ cannot contribute at all. Hence the required confinement to $H^0 \oplus H^2$ follows exactly when every contributing effective class satisfies $c_1 \cdot d \geq 0$, and it is not automatic: Iritani's theorem is stated for an arbitrary smooth projective X , and after several blowups in a weak factorization there is no reason for the intermediate ambient variety to have nef anticanonical class.

So the iterated receiver does not by itself remove arbitrary intermediate fourfolds from the problem. It removes them under a semipositivity hypothesis on the ambient, which the weak factorization does not supply. This is worth stating plainly, because it is the second proposed repair that fails on inspection, and it fails for a reason the draft would have inherited silently.

For the center the same analysis applies, with the additional and unchanged difficulty that its Novikov map is the noninjective one, which is why divisor tagging exists.

The scope consequence is the reason to try this first. If the ambient endpoint is handled this way, no inequality is needed for arbitrary intermediate fourfolds — the case that looked least

tractable. What remains is the center summands, and there the draft's own low-dimensional classification is restrictive: for nef canonical class it produces an integral- z gauge to a regular-singular connection, so there is no exponential separation at all and $w(\Delta\lambda) = 0$ again; the remaining minimal cases are $\mathbf{P}^1$, $\mathbf{P}^2$, ruled surfaces, and point blowups. Proving primitive-sixth vanishing directly for those, in the coordinates Iritani's specialization actually produces, is a smaller problem than a general transport theorem.

5 The framing route, which is probably the repair

Everything above treats the pro-Laurent bulk gauge of Lemma 4.5 as unavoidable. It may not be. Hinault, Yu, Zhang and Zhang work with F -bundles on the formal u -disc, where a framing is a regular flat connection and, in a framing-flat trivialization, the meromorphic connection takes the normal form

$$\nabla_{u\partial_u} = u\partial_u + u^{-1}K + (\mu D + H).$$

Their Theorem 4.34 characterizes when two connections of that shape are gauge-equivalent under

$$\Phi(u) \in \mathrm{GL}(H[[u]]), \quad \Phi_{ij}(u) \in k[c_1, \dots, c_r][[u]],$$

with Φ then unique given its value at $u = 0$; the conditions are that the leading operators K be conjugate by a parameter-polynomial ϕ , that the two μ agree, and that the diagonal entries of H agree after conjugation modulo the parameter ideal. Their applications are uniqueness of the decomposition for projective bundles and for blowups, complementing the existence theorems of Iritani–Koto and Iritani.

The shape of that gauge is exactly what this memo has been missing. A gauge in $\mathrm{GL}_n(K[[u]])$ has only nonnegative integral powers of the loop coordinate, adjoins no root of u , and is therefore fixed by the turn $u \mapsto e^{2\pi i}u$. Taking $\Gamma = 0$, so that $\mathcal{H} = \Omega_V((u))$ and Levelt–Turrittin is available, Theorem 7 applies with no further work:

$$\Phi \in \mathrm{GL}_n(K[[u]]), \quad \nabla' = \Phi \nabla \Phi^{-1} - \theta(\Phi) \Phi^{-1} \implies M_{f,V}(\nabla') \text{ and } M_{f,V}(\nabla) \text{ conjugate.}$$

No Hahn receiver, no product GP , and no inequality $ew(\Delta\lambda) < \varepsilon$. The scissors of Section 2 close because the gauge is regular rather than pro-Laurent: the entire obstruction analysed in Section 3 is a consequence of the draft's own choice to realize the bulk transport by a normalized gauge with unbounded negative loop order, not of anything in the comparison theorems.

What has to be checked is therefore a compatibility question, and it is much smaller than a transport theorem:

Does the decomposition used in Proposition 4.7, after the coefficient base changes and with $u = z$, coincide with an F -bundle isomorphism of the class Theorem 4.34 governs — that is, do the two connections being compared lie in $\mathcal{F}(H, \text{iso}, d, (c_i))$, and are its three conditions met?

Two remarks on scope. First, the identification is plausible on the authors' own terms, since their stated application is uniqueness of exactly the decompositions Proposition 4.7 imports for existence. Second, this does not touch the center specialization: their statement is about the decomposition as an F -bundle, upstream of the numerical center map, whose noninjectivity is what divisor tagging exists to repair. So the low-dimensional center problem survives intact,

while the frightening half — transporting framed monodromy across arbitrary intermediate fourfolds — would be removed.

If that bridge checks, Sections 2 to 3 remain accurate as an account of why the proof currently in the manuscript does not work, but they stop being the repair, and the manuscript should adopt the regular gauge instead of the pro-Laurent one wherever it can.

6 Four routes, in the order I would try them

1. **The framing theory of Hinault, Yu, Zhang and Zhang** (arXiv:2411.02266), whose reconstruction algorithm is the one Iritani–Koto added as their Section 5.8, and which is very likely the actual repair. See Section 5.
2. **The iterated receiver of Section 4**, now known to need a semipositivity hypothesis on the ambient variety that weak factorization does not supply. Still worth having where it applies, but it does not remove the arbitrary intermediate fourfolds.
3. **The low-dimensional centers directly**: primitive-sixth vanishing after Iritani’s specialization, using the classification the draft already proves, rather than through a general transport statement.
4. **Divisor tagging on its own terms**: whether the divisor equation gives a more explicit transport than the general positive-bulk gauge of Lemma 4.5. This lemma carries the same receiver problem and needs its own treatment either way.
5. **Normalize, then verify**. Replace “choose L so large” by the smallest weight satisfying separation, adopt the single-copy receiver of Section 3, and verify $ew(\Delta\lambda) < \varepsilon$ case by case.

Two further possibilities remain open rather than ordered. A transport theorem that never forms GP — arguing graded piece by graded piece, as the rank-one case does — escapes the criterion entirely, but by the scissors it must supply a canonical normal form over the receiver singling out the turn, and cannot be a pure valuation argument. And restricting Proposition 4.7 to centers satisfying the criterion is available, at a cost in generality that is the author’s call.

A separate scope reduction, also due to the same reader, would shrink what needs repairing: for the headline theorem the projective-bundle identity $\nu_6(X \times \mathbf{P}^1) = 2\nu_6(X)$ might be replaced by a direct quantum Künneth calculation, with $\nu_6(\mathbf{P}^4) = 0$ computed directly, leaving the blowup step for four-dimensional birational invariance as the only indispensable operation formula. The genus-eight corollary would then wait for the projective-bundle machinery.

7 Placement in the manuscript

Conditional on a receiver repair, and stated by label because these insertions renumber Section 4.

1. The ground field and solution algebra go immediately before “Apply Levelt–Turrittin after a ramification $z = w^e$ ”, which already refers to a universal formal solution algebra that is never constructed.

2. Lemma 4 and Proposition 6 go after `def:framed-sixth-multiplicity`, converting that definition into a characterization, with Remark 5 stating the hypothesis that makes it available.
3. Theorem 7 becomes `lem:frame-transport`, after the paragraph beginning “The frame matters” and before `def:pro-laurent-gauge-group`.
4. In `lem:pv-base-change`, the asserted final clause and the corresponding sentence of its proof become a reference to `lem:frame-transport` — but only once the receiver repair makes that lemma applicable there, since that lemma lives over the pro-Laurent receiver.
5. At (4.1c) in `lem:formal-base-shift`, one sentence identifying M^{bulk} as the transported framed automorphism, in the module frame, with the solution-frame statement cited.
6. `prop:framed-operations` cites the lemma at its three load-bearing steps and says which field plays the role of $\mathcal{H}$ at each, with the criterion verified there.
7. `lem:divisor-tagging` carries the same obstruction — it embeds the gauge of `lem:formal-base-shift`, whose negative z -order is unbounded, together with every transported operator, into one ordered Hahn field — and needs whatever repair the receiver gets.

Mechanically, a new theorem-like environment must carry exactly one semantic label and a matching row in the claim map consumed by the manuscript’s source-only gate, with an empty declaration list exactly when its coverage is recorded as absent, and the coverage snapshot string in both verification READMEs must be regenerated.

8 Corrections to the first version of this memo

For readers of the first version, these are the substantive changes.

1. The Constants Lemma was stated without the constant-coefficient hypothesis and is false without it; see Remark 5. The framed-operator characterization inherits the hypothesis.
2. The scissors were only implicit. The transport theorem has no verified instance in the application, and that now appears where the theorem does.
3. The claim that the criterion is scale-invariant, and the inference that the separating parameter L cannot be tuned, were both wrong. Tuning L down is now the first proposed route.
4. The cubic-endpoint bullet compared two different normalizations. Corrected above.
5. The finite-tensor objection to the receiver was wrong twice over: the pro-Laurent gauge lies in the Hahn factor alone, and the splitting gauge’s coefficients lie in one finite extension of $H_{0,j}$. It is replaced by the gauge-relation argument.
6. The remark that a tensor product of two fields is not a domain was asserted as a general principle, which is false. The correct statement is that $\mathcal{R}_j$ need not be one.

7. The obstruction is now given in its order-free form, which answers two of the questions the first version left open: no completion helps, and the criterion is not an artifact of the ordered receiver.
8. Smaller repairs: the freeness proof (the relevant quotient group has torsion), the commutation of σ with the derivation, the inverse of M , the range of the exponents, the ramification factor in the splitting rate, and the divisor-tagging observation, which is new.

9 Questions

1. Do the two connections compared in Proposition 4.7 lie in the class Theorem 4.34 of Hinault–Yu–Zhang–Zhang governs, and are its three conditions met? This is the question that decides the repair.
2. For the low-dimensional centers, can primitive-sixth vanishing be proved directly in the coordinates Iritani’s specialization produces, given that the nef-canonical case already yields an integral- z gauge to a regular-singular connection?
3. Is there a canonical normal form over the receiver singling out the turn, which is what a transport argument avoiding GP needs and what the existence of a Picard–Vessiot ring does not give?
4. Failing all of these, what is the right restriction on centers, and does weak factorization stay inside it?